

Vitor Negromonte Cabral de Oliveira

Last update: January 5, 2026

Education

Federal University of Pernambuco

B.Sc Information Systems

Recife

2022–2027

Major in Statistics from 2022 to 2025 then transferred to Info. Syst.

Activities: AI League co-founder, Teaching Assistant (Introduction to Deep Learning), Research Assistant (INES, Geraia).

Work Experience

Machine Learning Engineer

Freelancer

Jan 2025 – Sep 2025

Remote

Designed and deployed a full-stack computer vision pipeline for agricultural applications. Managed data collection, dataset creation, image preprocessing, model training, and real-time edge inference.

Stack: Python (PyTorch, OpenCV), Docker, Linux, Shell, Git.

Data Scientist

redduo.ai

Nov 2023 – Jul 2024

Recife, BR

Supported business intelligence projects through data analysis and automation tool development. Developed and optimized an NLP pipeline, improving model performance and enabling scalable deployment across production environments.

Stack: Python (Scikit-learn, PyTorch), AWS (Lambda, Cognito), Power BI.

Research Experience

California Institute of Technology + Federal University of Pernambuco

Undergraduate Research Assistant

Apr 2025 – Aug 2025

Remote

Group: CAMBIA. Advisors: Prof. Tsang Ing Ren (UFPE), Dr. Alexandre Cunha (Caltech) Conducted research on techniques for image segmentation, with a focus on biomedical imaging.

GERAIA

Member

Nov 2023 – Today

Recife, BR

Conducted research in Generative AI with a focus on evaluating language models in Portuguese, assessing their performance and adaptability to emergent linguistic variations

National Institute of Science and Technology in Software Engineering (INES)

Undergraduate Research Assistant

Aug 2023 – May 2024

Remote/Recife, BR

Assisted in the development of quantitative analysis tools and contributed to improving accessibility features in mobile applications aimed at supporting adults on the autism spectrum. Published at SBSI – 2025 and IHC 2024.

Extracurricular

Co-organizer

II Advanced Summer School on Responsible AutoML

Nov 2025

Recife, BR

Event made by the Center for Informatics (UFPE) and the Engineering Faculty of the University of Porto. Coordinated outreach and partnership with interested organizations.

Co-founder & Outreach Director

Ligia – Federal University of Pernambuco's Artificial Intelligence Academic League

Mar 2024 – Today

Recife, BR

Lead partnership development, organize AI-focused events, and create educational content to promote AI learning and application across multiple disciplines. The league is a non-profit initiative affiliated with the CIn.AI research group at the Federal University of Pernambuco.

Projects

F1Predict: Formula 1 Race Prediction System –

Built an end-to-end system for predicting Formula 1 race outcomes, including data collection, feature engineering, model training, and deployment. Developed a REST API with FastAPI and PostgreSQL for data management and model serving. Achieved a 1-second prediction error, demonstrating high real-time accuracy. **Tools:** Python, PyTorch, scikit-learn, FastAPI, PostgreSQL.

MARS: Multi Agent Recommendation System –

Developed a FastAPI-based system that automates research paper collection from ArXiv using LLM agents to review and filter relevant papers. **Tools:** Python (FastAPI).

Parkinson Diagnosis using Computer Vision: Campus Party'24 Keynote –

Created a CNN model for Parkinson's disease detection using spiral-drawing images. Outperformed state-of-the-art methods by $\approx 10\%$, achieving 95% accuracy on the benchmark dataset. **Tools:** Python, PyTorch, OpenCV, Optuna, Zeus.

Computer Vision in Breast Cancer Diagnosis: A Comparative Study with CBIS-DDSM Data –

Performed a comparative study of multiple CNN architectures for early breast cancer detection using mammography images. **Tools:** Python, TensorFlow, OpenCV.

Publications

- [2] D. M. Ribeiro, F. V. Melo, **V. Negromonte**, C. Pereira, A. P. C. Steinmacher, K. Gama, "A Mapping Review to Understand Web and Mobile Apps Accessibility for Adults with Autism", SBSI – 2025
- [1] D. M. Ribeiro, F. V. Melo, **V. Negromonte**, G. W. Matias, A. Farias, C. Azul, A. P. Chaves, K. Gama, "A Comparative Study on Accessibility for Autistic Individuals with Urban Mobility Apps", IHC – 2024

Awards & Grants

Cohere Labs Catalyst Grants Program	2025
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Teaching

Federal University of Pernambuco

IF866 – Introduction to Deep Learning, Head TA	2025 – Today
IF866 – Introduction to Deep Learning	2024 – 2025
IF866 – Computational Creativity	2024 – 2025

Skills & Tools

Programming – Python, R, SQL, $\LaTeX$, Bash.

Data Science & ML – PyTorch, NumPy, Pandas, Matplotlib, Seaborn, SKLearn, Weights&Biases, MLFlow, HuggingFace.

SWE & DevOps – FastAPI, Flask, Docker, Docker Compose, AWS (Lambda, Cognito), PostgreSQL.

Languages – Portuguese (native), English (C1).

Interests – Explainable AI, Knowledge Representation, Meta-Learning, Data-centric modeling, Neural Algorithmic Reasoning.